

Background

Quadruped robots offer superior mobility in complex and unstructured environments. However, deployment is limited by the need for specialized control and technical expertise. Traditional interfaces are not intuitive for non-expert users. Recent advances in Large Language Models (LLMs) enable natural language interaction with robotic systems. This project integrates speech recognition, perception, and navigation to enable intuitive human-robot interaction.



Objectives

- Enable natural language control of quadruped robots through LLM-based planning and task decomposition built on top of ROS2 with Unitree Go2 hardware.
- Integrate 3D scene understanding (Gaussian Splatting) with semantic reasoning (LangSplat V2) for spatial queries.
- Develop modular and interpretable agentic control using LangGraph state machines and tool-based execution with voice control support.
- Train navigation policies in Habitat-Lab simulation for safe, scalable policy development.

Methods Overview

Environment Mapping

- **Gaussian Splatting:** Constructs 3D point cloud from video as Gaussian primitives with position, covariance, and color parameters.
- **LangSplat V2:** Embeds CLIP language features into primitives, enabling semantic queries that return 3D coordinates.

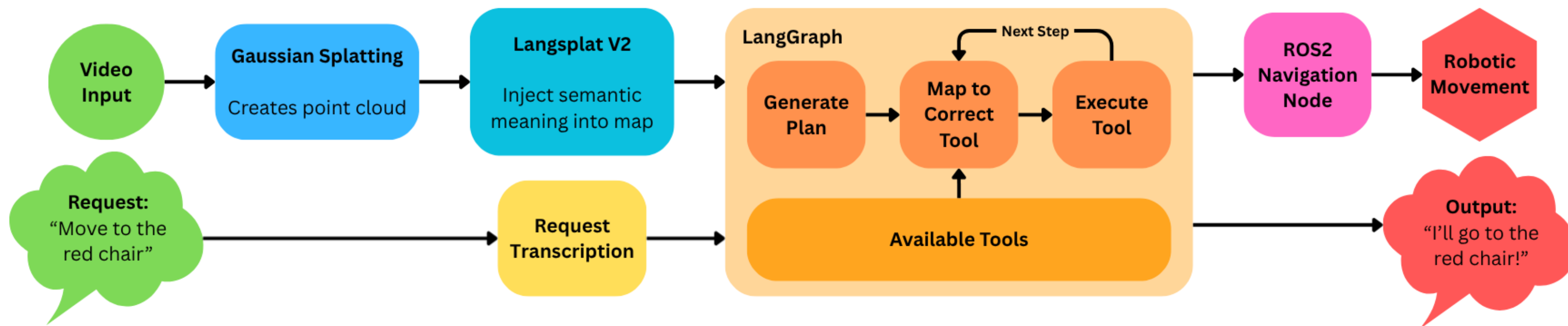
Agentic Control (LLM-Nav)

- **LangGraph State Machine:** *Plan* → *Execute Step* → *Run Tools* → *Advance* loop decomposes natural language commands into executable tool sequences, querying LangSplat for object coordinates to generate motion plans using GPT-4o-mini.
- **ROS2 Hardware Interface:** Tool calls execute via Go2-ROS2-SDK, leveraging built-in SLAM for real-time localization and spatial awareness during navigation.

Reinforcement Learning (RL)

- **Habitat-Lab Simulation:** Train navigation policies using reward-based learning in photorealistic environments for safe, scalable policy development.

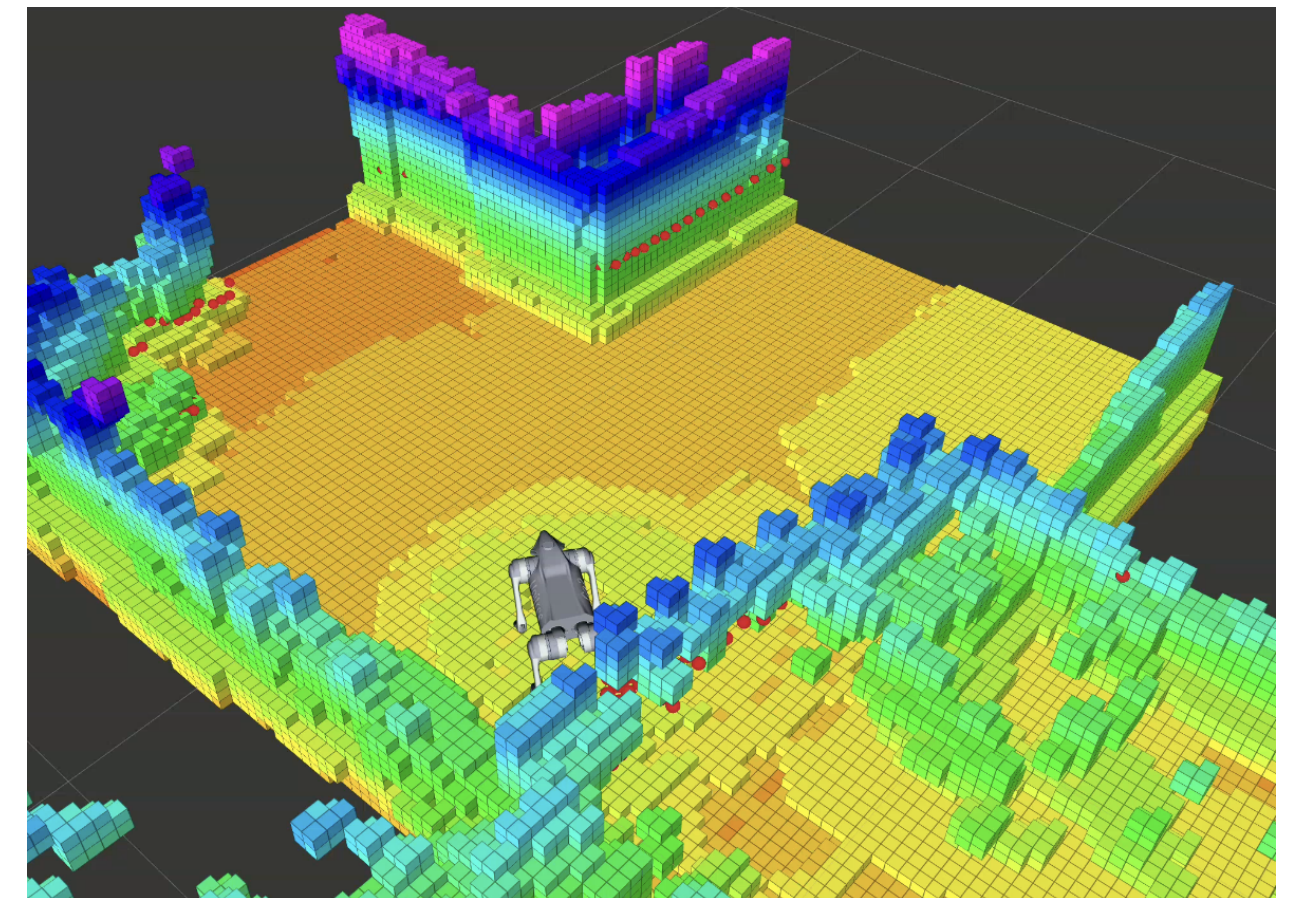
LLM Navigation Pipeline



Voice commands are transcribed and processed by a LangGraph agent that queries a semantic 3D map for object coordinates, then generates and executes motion plans via ROS2.

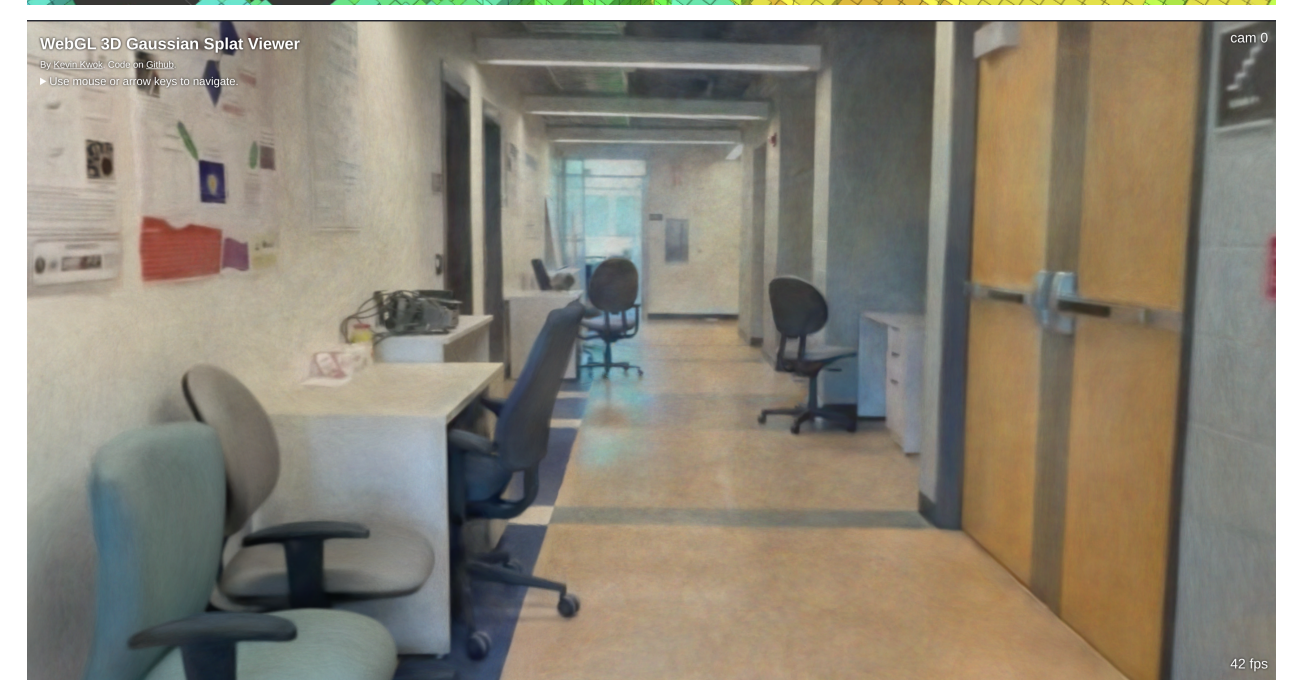
Mapping with Gaussian Splatting [1] & Langsplat V2 [2]

- **3D Scene Reconstruction:** Gaussian Splatting models the environment as a collection of 3D Gaussian primitives, each defined by position (μ), covariance (σ), opacity (α), and color, creating high-fidelity spatial maps from video.
- **Semantic Integration:** LangSplat V2 embeds CLIP language features into each Gaussian primitive, enabling the map to understand natural language object queries.
- **Spatial Query Interface:** The system returns 3D coordinates (x, y, z) when queried with commands like "where is the red chair?", bridging semantic understanding with actionable navigation targets.



Reinforcement Learning (RL)

- **Habitat-Lab Simulation:** Trained navigation policies for virtual Go2 in photorealistic environments using reward-based learning to optimize task completion and obstacle avoidance.
- **Sim-to-Real Pipeline:** Established foundation for transferring learned behaviors from simulation to physical hardware deployment.



SLAM via Go2-ROS2-SDK

- **ROS2 Interface & Localization:** Leverage Go2-ROS2-SDK's built-in SLAM capabilities for real-time localization and mapping, providing spatial awareness for autonomous navigation.

References

- [1] Bernhard Kerbl et al. "3d gaussian splatting for real-time radiance field rendering." In: *ACM Trans. Graph.* 42.4 (2023), pp. 139–1.
- [2] Wanhua Li et al. "Langsplatv2: High-dimensional 3d language gaussian splatting with 450+ fps". In: *arXiv preprint arXiv:2507.07136* (2025).

Impact

- **Accessibility:** Natural language control eliminates the need for specialized training, making advanced robotics accessible to non-experts through intuitive voice commands.
- **Safety:** Simulation-based RL training in Habitat-Lab enables safe policy development without risk to hardware or people.
- **Applications:** Home assistance, elder care, warehouse automation, search and rescue, and collaborative robotics in unstructured environments.

Acknowledgment

- **Dr. Jonathan Shihao Ji** for his tremendous support throughout this project's development. We greatly appreciate his assistance.
- **Go2-ROS2-SDK, Habitat-Lab, Gaussian Splatting, Langsplat V2, and LangGraph** researchers, maintainers, and developers.